ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE C

MARKS: 80 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET second paper (drawing practical, 80 marks): redraw a regulated 5 V power supply, draw the circuit diagram of a breadboard layout, and redraw and build an LM358 light comparator and a 555 one-shot timer in draw.io and Fritzing.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.
2. Use appropriate application programs: draw.io (or similar) for circuit diagrams and Fritzing for breadboard layouts. If you do not have a computer, draw neatly by hand.
3. Hand in only ONE printout (or page) per question. Every printout must show the question number and your name or examination number.
4. Save your work regularly.

QUESTION 1

Redraw the circuit diagram of the regulated 5 V power supply below using a software tool such as draw.io.

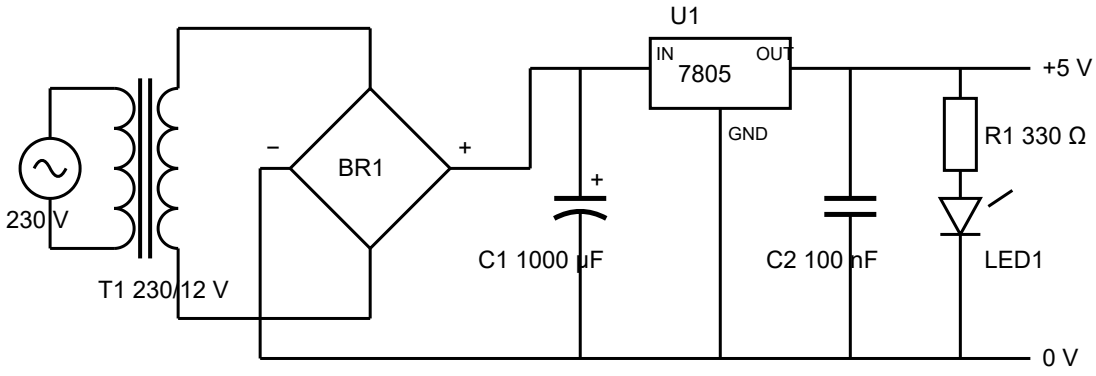


FIGURE 1: Regulated 5 V power supply

Redraw FIGURE 1 using draw.io (or neatly by hand), with the correct symbols and labels. (10)

[10]

QUESTION 2

A technician built the transistor circuit described in the table below on a breadboard. Draw the CIRCUIT DIAGRAM of this layout using a software tool such as draw.io.

Component	Connections on the breadboard
9 V battery	+ to the top (+) rail; − to the bottom (−) rail
LDR	from the + rail to row 10
10 kΩ potentiometer	one outer pin and the wiper in row 10; other outer pin in the − rail
2,2 kΩ resistor	from row 10 to row 15
BC547 transistor	collector row 20, base row 15, emitter in the − rail
Active buzzer	+ pin in the + rail, − pin in row 20

1N4148 diode	cathode in the + rail, anode in row 20

Draw the circuit diagram of the breadboard layout described above. (10)

[10]

QUESTION 3

3.1 Redraw the circuit diagram of the light-operated comparator below using draw.io. The LED comes on when the light level falls below the level set by VR1. (15)

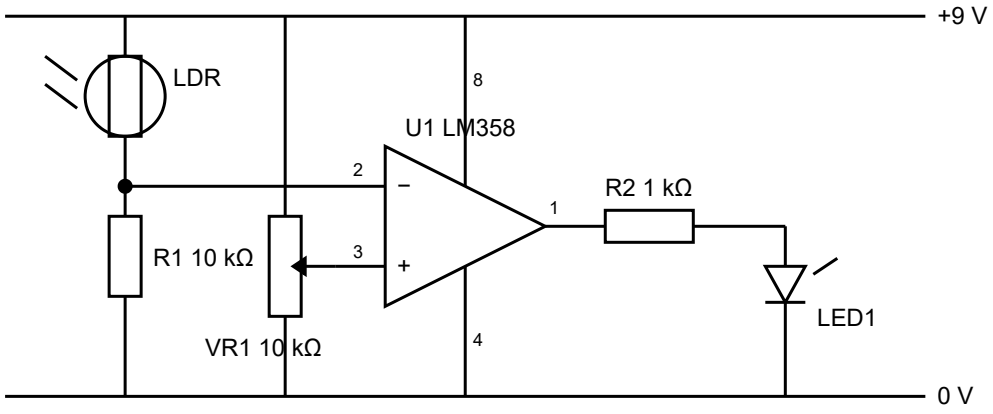


FIGURE 2: LM358 light comparator (lines that cross without a dot are not connected)

3.2 Using Fritzing, create the breadboard layout of the comparator circuit in FIGURE 2. (15)

[30]

QUESTION 4

4.1 Redraw the circuit diagram of the 555 monostable (one-shot) timer below using draw.io. When the button is pressed, the LED stays on for about 5 seconds. (15)

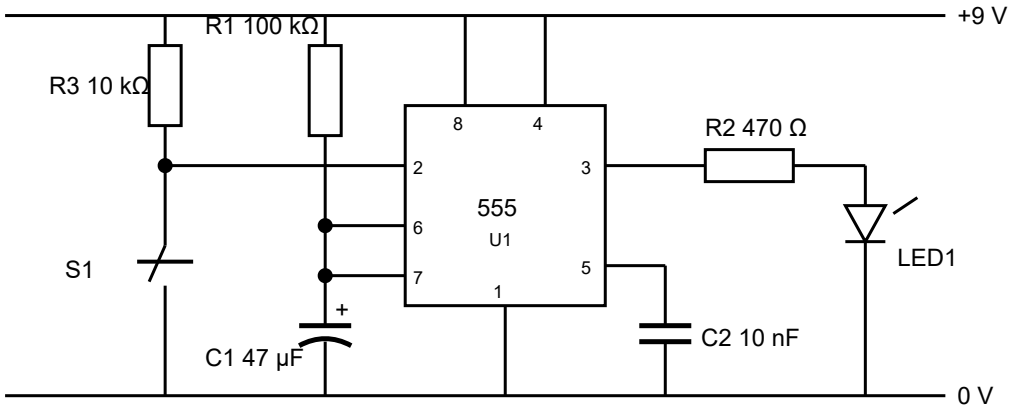


FIGURE 3: 555 monostable timer (lines that cross without a dot are not connected)

4.2 Using Fritzing, create the breadboard layout of the 555 monostable timer in FIGURE 3. (15)

[30]

TOTAL: 80

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END OF QUESTION PAPER